Problem · Topic 3 · 4 points · 60 min

P14 — AVIEBRO: a half-year close with a volume discount, a mortgage and prior losses

AVIÓNICA DEL EBRO S.L. (AVIEBRO), a Zaragoza workshop dedicated to the repair and calibration of avionics equipment for regional airlines and flying schools, must present to the bank that finances it the financial statements (balance sheet and Income Statement) as of 30 June 2020 (30/06/2020). The financial director has the balance of 31 December 2019 (31/12/2019), included below, and the following operations of the first semester, from 1 January 2020 (01/01/2020) to 30 June 2020 (30/06/2020):

I. *The company has invoiced repairs for 1.600 m.u. every month. The payment condition agreed with the customers is payable to 60 days. A regional airline, which represents 40% of the invoicing of every month, surpassed in March the annual volume agreed in its contract, and from 1 April 2020 (01/04/2020) it receives a discount of 5% on all its invoices.*

II. *The customer debt outstanding at 31 December 2019 (31/12/2019) was charged in January, except 250 m.u. from a flying school that has been granted to pay it on 30 September 2020 (30/09/2020). The suppliers of the previous year were paid in February. As the company closed 2019 with losses, there were no taxes to pay.*

III. *At 31 December 2019 (31/12/2019) there is a stock (stock in stock) of 300 spare parts valued at 2 m.u. / unit. On 1 February 2020 (01/02/2020) the company purchased 400 parts at 2 m.u., 50% paid in cash and 50% payable to 90 days; on 1 May 2020 (01/05/2020) it purchased 300 parts more at the same price, payable to 120 days. At 30 June 2020 (30/06/2020) there remain 250 parts in the warehouse.*

IV. *The technicians cost 600 m.u. / month, paid in the same month, to which 180 m.u. / month of Social Security must be added, paid the month after. The overheads (electricity, telephone, cleaning) are 100 m.u. / month and are paid two months after being made. On 1 March 2020 (01/03/2020) the company carried out a renting contract for a test bench of 120 m.u. monthly, paid at the end of each month.*

V. *The patent of the balance (a calibration procedure) is amortized linearly in 5 years with residual value zero; the computer equipment was bought on 1 January 2019 (01/01/2019) and is amortized linearly in 3 years with residual value zero.*

VI. *On 1 April 2020 (01/04/2020) AVIEBRO bought a small hangar next to the airport: the building is valued at 2.000 m.u. and the land at 1.000 m.u. It paid 20% in cash and the rest with a mortgage loan at 4% annual interest, to be returned in installments of 120 m.u. every 1 October and every 1 April, the first one on 1 October 2020 (01/10/2020); the same days it pays the interest of the semester calculated on the debt at the beginning of the semester. The building is amortized linearly in 50 years with residual value zero.*

VII. *The long-term loan of the balance bears 5% annual interest, which is paid every 31 December, and it is returned in one single payment on 31 December 2022 (31/12/2022).*

VIII. *The corporation tax is 25%. The taxable base of the semester is the profit before taxes minus the losses from previous years still pending to offset, and the resulting tax will be paid on 20 July 2020 (20/07/2020) (counted in the balance as "taxes to pay"). No dividends are distributed at the half-year.*

BALANCE 31 December 2019 (31/12/2019)

ASSETS	***m.u.***	***LIABILITIES***	***m.u.***
NON-CURRENT ASSETS	***1.900***	***OWNERS EQUITY***	***2.700***
Patent	*1.200*	*Paid-in capital*	*3.200*
Accum. amortization of the patent	*(300)*	*Losses from previous years*	*(500)*
Computer equipment	*900*	***NON-CURRENT LIABILITIES***	***1.000***
Accum. depreciation of computer equipment	*(300)*	*Long-term loan (31/12/2022)*	*1.000*
Financial shares	*400*	***CURRENT LIABILITIES***	***500***
CURRENT ASSETS	***2.300***	*Suppliers*	*500*
Stock of spare parts	*600*		
Accounts receivable	*1.000*		
Cash	*700*		
Total Assets	***4.200***	***Total Liabilities + OE***	***4.200***

Required:

1. *Deduce the exact date on which the patent was acquired, and construct the Income Statement (Profit and Loss Account) of AVIEBRO for the first semester of 2020, explaining the intermediate calculations.*
2. *Calculate the cash and construct the balance sheet at 30 June 2020 (30/06/2020).*
3. *Determine the values of the ROA and ROE ratios of the semester, and indicate how the Income Statement and the balance sheet would change if the tax authority did not allow to offset the losses from previous years.*
4. *Instead of buying the hangar, the company could have rented it from 1 April 2020 (01/04/2020) for 50 m.u. monthly, paid at the end of each month. Indicate the changes in the Income Statement, the cash and the balance sheet at 30 June 2020 (30/06/2020). In which case is the profit higher, and in which case is the final cash balance higher? Discuss, reasoned, why.*

Setup

Time window: six months, 01/01/2020–30/06/2020. Depreciation and interest are prorated by months; expenses are recorded when incurred even if they are paid later [Slide 8].

Hypotheses:

1. The airline's discount applies to the invoices of April, May and June only.
2. "Payable to 90 days" from 1 February means payment on 1 May; "payable to 120 days" from 1 May means payment at the end of August, outside the semester.
3. The losses from previous years are offset for tax purposes, but they stay in the balance sheet until the annual meeting distributes the profit of the year; the result of the semester is shown separately in equity.
4. Land is not depreciated.

Extension (not defined in the course decks). $\mathrm{ROA} = \text{Income from operations}/\text{Total assets}$ and $\mathrm{ROE} = \text{Net profit}/\text{Owners equity}$, both for the semester and on the balance at 30/06/2020.

Part (a) · 1.25 points

Deduce the purchase date of the patent and prepare the Income Statement of the semester

Step 1. Purchase date of the patent

$$D_{\text{patent}} = \frac{1.200}{5} = 240 \text{ m.u./year}, \qquad \frac{300}{240} = 1{,}25 \text{ years} = 15 \text{ months before } 31/12/2019 \Rightarrow \text{1 October 2018 } (01/10/2018)$$

Step 2. Revenue

$$\text{Discount per month} = 0{,}05 \times 0{,}40 \times 1.600 = 32 \text{ m.u.}, \qquad \text{Revenue} = 6 \times 1.600 - 3 \times 32 = 9.600 - 96 = 9.504 \text{ m.u.}$$

Step 3. Parts consumed

$$\text{Consumption} = 600 + 400 \times 2 + 300 \times 2 - 250 \times 2 = 600 + 800 + 600 - 500 = 1.500 \text{ m.u.}$$

Step 4. Personnel, overheads and renting

$$\text{Personnel} = 6 \times (600 + 180) = 4.680, \qquad \text{Overheads} = 6 \times 100 = 600, \qquad \text{Renting} = 4 \times 120 = 480 \text{ m.u.}$$

Step 5. Depreciation of the semester

$$D_{\text{patent}} = 240 \times \frac{6}{12} = 120, \qquad D_{\text{computer}} = \frac{900}{3} \times \frac{6}{12} = 150, \qquad D_{\text{building}} = \frac{2.000}{50} \times \frac{3}{12} = 10$$

$$D = 120 + 150 + 10 = 280 \text{ m.u.}$$

Check of the computer equipment: one year at 300 m.u. gives the 300 of the 2019 balance (checked).

Step 6. Interest of the semester (accrued, nothing paid)

$$I_{\text{mortgage}} = 2.400 \times 0{,}04 \times \frac{3}{12} = 24, \qquad I_{\text{loan}} = 1.000 \times 0{,}05 \times \frac{6}{12} = 25 \text{ m.u.}$$

The mortgage is $3.000 - 0{,}20 \times 3.000 = 2.400$ m.u.

Step 7. Tax

$$\text{Taxable base} = 1.915 - 500 = 1.415, \qquad \text{Tax} = 0{,}25 \times 1.415 = 353{,}75 \text{ m.u.}$$

Step 8. Income Statement 01/01/2020–30/06/2020

Line	Computation	m.u.
Repair revenue	Step 2	9.504
– Spare parts consumed	Step 3	1.500
– Personnel (wages and Social Security)	Step 4	4.680
– Overheads	Step 4	600
– Renting of the test bench	Step 4	480
– Depreciation and amortization	Step 5	280

Line	Computation	m.u.
= Income from operations	$9.504 - 7.540$	**1.964**
– Financial expenses	$24 + 25$	49
= Profit before taxes	$1.964 - 49$	**1.915**
– Corporation tax	Step 7	353,75
= Net profit of the semester	$1.915 - 353{,}75$	**1.561,25**

KEY RESULTS

- Patent bought on 1 October 2018 (01/10/2018).
- Revenue 9.504 (discount 96); parts consumed 1.500; depreciation 280 m.u.
- Income from operations 1.964; interest 49; profit before taxes 1.915 m.u.
- Taxable base 1.415; tax 353,75; net profit 1.561,25 m.u.

Part (b) · 1.25 points

Compute the cash and prepare the balance sheet at 30 June 2020

Step 1. Cash account of the semester

Inflows	m.u.	Outflows	m.u.
2019 receivables ($1.000 - 250$)	750	Suppliers 2019	500
Invoices January–March (3×1.600)	4.800	Parts of 01/02 ($400 + 400$)	800
Invoice of April ($1.600 - 32$)	1.568	Technicians' wages	3.600
Mortgage loan	2.400	Social Security, January–May	900
		Overheads, January–April	400
		Renting, March–June	480
		Hangar (building and land)	3.000
Total	**9.518**	**Total**	**9.680**

$$\text{Cash}_{30/06/2020} = 700 + 9.518 - 9.680 = 538 \text{ m.u.}$$

Step 2. Items still open at 30/06/2020

- Receivables: May and June invoices, $2 \times (1.600 - 32) = 3.136$, plus the 250 deferred to 30/09/2020: 3.386 m.u.
- Suppliers: the parts of 01/05 ($300 \times 2 = 600$), due at the end of August.
- Mortgage: installments of 01/10/2020 and 01/04/2021 ($2 \times 120 = 240$) are current; $2.400 - 240 = 2.160$ non-current.

Step 3. Balance sheet at 30/06/2020

Assets	**m.u.**	**Liabilities and owners equity**	**m.u.**
Patent (1.200 − 420)	780	Paid-in capital	3.200
Computer equipment (900 − 450)	450	Losses from previous years	(500)
Land	1.000	Profit of the semester	1.561,25
Building (2.000 − 10)	1.990	**Owners equity**	**4.261,25**
Financial shares	400	Long-term loan (31/12/2022)	1.000
Non-current assets	**4.620**	Mortgage loan, non-current part	2.160
Stock of spare parts (250 × 2)	500	**Non-current liabilities**	**3.160**
Accounts receivable	3.386	Mortgage loan, installments within one year	240
Cash	538	Interest payable (24 + 25)	49
		Suppliers	600
		Social Security payable	180
		Overheads payable	200
		Taxes to pay	353,75
Current assets	**4.424**	**Current liabilities**	**1.622,75**
Total assets	**9.044**	**Total liabilities and equity**	**9.044**

KEY RESULTS

- Cash at 30/06/2020: 538 m.u.
- Receivables 3.386; non-current assets 4.620; total 9.044 m.u.
- Owners equity 4.261,25; non-current liabilities 3.160; current liabilities 1.622,75 m.u.

Part (c) · 0.75 points

Compute ROA and ROE of the semester, and the effect of not offsetting the prior losses

Step 1. Ratios of the semester (extension)

$$\text{ROA} = \frac{1.964}{9.044} = 21{,}72\%, \qquad \text{ROE} = \frac{1.561{,}25}{4.261{,}25} = 36{,}64\%$$

Step 2. Without the offset of the losses

$$\text{Tax} = 0{,}25 \times 1.915 = 478{,}75\ (+125), \qquad \text{Net profit} = 1.915 - 478{,}75 = 1.436{,}25\ (-125)$$

In the balance, "taxes to pay" rises by 125 m.u. and the profit of the semester (equity) falls by 125 m.u.; total assets and cash do not change, because the tax is paid only in July. The ROE falls to $1.436{,}25/4.136{,}25 = 34{,}72\%$. The offset is worth exactly $0{,}25 \times 500 = 125$ m.u. of tax saved.

KEY RESULTS

- ROA = 21,72%; ROE = 36,64% (semester, extension).
- Without the offset: tax 478,75 (+125); net profit 1.436,25 (–125); taxes to pay +125; equity 4.136,25; totals unchanged; ROE 34,72%.

Part (d) · 0.75 points

Renting the hangar instead of buying it; which case gives more profit and which more cash?

Step 1. New results

$$\text{Income from operations} = 1.964 + 10 - 3 \times 50 = 1.824, \qquad \text{Profit before taxes} = 1.824 - 25 = 1.799$$

$$\text{Tax} = 0{,}25 \times (1.799 - 500) = 324{,}75, \qquad \text{Net profit} = 1.799 - 324{,}75 = 1.474{,}25$$

Step 2. New cash

$$\text{Cash} = 538 + 600 - 3 \times 50 = 988 \text{ m.u.}$$

Step 3. Changes in the statements

Item	Buying	Renting	Change
Depreciation	280	270	−10
Rent of the hangar	0	150	+150
Financial expenses	49	25	−24
Net profit	1.561,25	1.474,25	−87
Land and building (net)	2.990	0	−2.990
Cash	538	988	+450
Total assets	**9.044**	**6.504**	**−2.540**
Owners equity	4.261,25	4.174,25	−87
Mortgage loan (current and non-current)	2.400	0	−2.400
Interest payable	49	25	−24
Taxes to pay	353,75	324,75	−29
Total liabilities and equity	**9.044**	**6.504**	**−2.540**

Discussion. The **profit is higher buying the hangar**: three months of rent (150) cost more than the depreciation of the building (10) and the mortgage interest (24) together, because the land is not depreciated and the mortgage is cheap. The **cash is higher renting**: the company keeps the 600 m.u. of down payment and pays only 150 of rent. Buying gives more profit but ties up cash and adds 2.400 m.u. of debt; renting preserves liquidity, which matters for a workshop that ends the semester with only 538 m.u. in the bank.

KEY RESULTS

- Income from operations 1.824 (−140); interest 25; tax 324,75; net profit 1.474,25 (−87).
- Cash 988 (+450); total assets 6.504; no mortgage.
- Profit higher buying; cash higher renting.

Interpretation and checks

- **Dimensional check:** m.u./month × months gives m.u.; rates × m.u. × months/12 give m.u.; ratios dimensionless.
- **Half-year ratios:** ROA and ROE of a semester are not annual returns; annualising them would roughly double them.

- **Profit is not cash:** the company earned 1.561,25 m.u. but its cash fell from 700 to 538 m.u.: the hangar, the uncollected May–June invoices and the growth of receivables absorbed it.
- **Typical mistakes:** depreciating the land, applying the airline discount to the whole semester, paying the mortgage interest in the semester, and taxing the profit without offsetting the losses as the statement allows.